MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5R4GA 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Rectifier, full wave, vacuum

Outline	---	See figure 1
Base	---	B5-15 (phenolic)
Envelope	---	T12
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	4	6	8
Element	---	nc	f	2a	1a	f

ABSOLUTE MAXIMUM RATINGS: See notes 1, 2 and 3

Parameter:	Ef	Epp/a	epx	ib/a	Io	Rg	Ck	Zp/a	tk
Unit:	Vac	Vac	vac	ma	mAdc	Ohms	µF	Ohms	sec
Maximum:	5.5	---	3,050	---	0	---	4	575	0
	5.5	---	2,900	Note 4	190	---	---	---	10
	5.5	---	2,800	---	165	---	---	575	10
	5.5	---	2,300	Note 5	275	---	---	---	10
	5.5	---	2,150	700	275	---	---	---	10
Minimum:	4.5	---	---	---	---	---	---	---	---
TEST CONDITIONS:	5.0	850	---	---	---	3,500	4	Note 6	---

GENERAL:

Qualification - Not required

1/ Formerly type 5R4GYA.

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1							
1201	Short and discontinuity detection		0.4	II	---	---	---	---
1231	Emission (1)	E2b=0; E1b=75 Vdc	0.65	II	I1s	225	400	mAdc
1231	Emission (2)	E1b=0; E2b=75 Vdc	0.65	II	I2s	225	400	mAdc
1353	Operation of rectifiers(1)	Full wave; tk=10 See note 7	0.65	II	Io	245	---	mAdc
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		4.0	S3	---	---	---	---
1301	Filament current		---	---	If	1.8	2.2	A
1353	Operation of rectifiers(2)	Full wave; Eapx=2,800 vac; Zp/a = 500 ohms; RL = 7,000 ohms; tk = 10	---	---	Io	140	---	mAdc
---	Alinement	See note 8	---	---	---	---	---	---
1002	Barometric pressure, reduced	Pressure = 140 ±10 mmHg; Ef = 5.0 V; Epp/a = 950 Vrms; CL = 4 µF; RL/Io = 165 mA; full wave; Zp/a = 200 ohms (including transformer) See note 9	---	---	---	---	---	---
1031	Low-frequency vibration	No voltages	---	---	---	---	---	---
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
1111	Base pin solder depth		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	Group B; Epp=750 Vac; Io=275 mAdc; full wave	---	---	t	500	---	hrs
---	Life test end-point:							
1353	Operation of rectifiers(1)	Full wave; tk = 10	---	---	Io	210	---	mAdc

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NOTES:

1. The peak inverse anode voltage, epx (no load) shall be 2,800 volts (maximum) for TUT operation at 20,000 feet (maximum). Epx (no load) shall be 2,400 volts (maximum) for TUT operation at 40,000 feet (maximum).

2. The TUT shall be operated in a vertical position. If the TUT is mounted in a horizontal position, pins 1 and 4 shall be in a vertical plane.

3. Satisfactory operation of the TUT under conditions within area I of figure 2, may be obtained without preheating the filament. For satisfactory operation of the TUT under conditions within area II of figure 2, preheat the filament for 10 seconds prior to applying anode voltage.

4. The input choke filter shall be 10 henries (minimum) for a 60 Hz power supply.

5. The input choke filter shall be 5 henries (minimum) for a 60 Hz power supply.

6. The TUT anode impedance shall not be greater than 200 ohms.

7. This test shall be performed at the conclusion of the holding period.

8. The plane of the TUT filament shall be parallel to the plane passing through the centerline of pins 1 and 4 within ±22.5 degrees.

9. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

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Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 99
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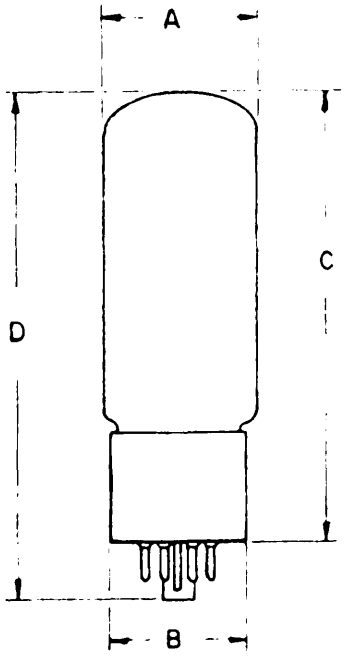
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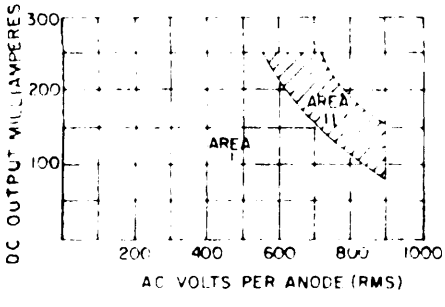
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
A	1.438 (36.53)	1.562 (39.67)
B		1.375 (34.93)
C	4.000 (101.60)	4.375 (111.13)
Quality conformance inspection, part 2		
D		4.938 (125.43)

FIGURE 1. Outline drawing of electron tube type 5R4GA.



Area I. Filament and anode voltages may be applied simultaneously.

Area II. Filament should be allowed to reach operating temperature before anode voltage is applied. For average conditions, the delay is approximately 10 seconds.

FIGURE 2. Operating characteristics.

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MIL-E-1/344C.	ELECTRON TUBE, RECEIVING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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